

EECE 568 - Assignment 1

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Due Date: October 1, 2026 - 11:59 pm

Late submissions will not be accepted. In case of any medical or family emergency, please inform the instructors.

Deliverables: Please upload a ZIP file to Canvas containing a PDF for section 1 and a Jupyter Notebook (.ipynb) for section 2.

1 Math Review and Theoretical Questions (50 marks)

1.1 Eigenvalue (6 marks)

For Matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$ Prove:

- If $m = n$, then A and A^T have the same set of eigenvalues
- AB and BA have the same set of eigenvalues

1.2 Rank (3 marks)

Prove that if $A \in \mathbb{R}^n$ is a full rank matrix, matrices B and $A^{-1}BA$ have the same set of eigenvalues.

1.3 Definite Matrix (3 marks)

A symmetric matrix $A \in \mathbb{R}^n$ is positive definite if $x^T Ax > 0$ for all non-zero vectors $x \in \mathbb{R}^n$. Prove if matrix A is positive definite, it is full rank and invertible.

1.4 Gram Matrix (6 marks)

Given any matrix $A \in \mathbb{R}^{m \times n}$, the matrix $G = A^T A$ is called a Gram matrix

Prove:

- G is always positive semidefinite
- If $m \geq n$ and A is full rank, then G is a positive definite matrix

1.5 Matrix Derivative (6 marks)

Consider the following function:

$$f(x, y) = \begin{bmatrix} f_1(x, y) \\ f_2(x, y) \\ f_3(x, y) \end{bmatrix} = \begin{bmatrix} x^2 y^3 \\ 4x^2 + \cos(y) \\ 4y^2 - 2x^2 \end{bmatrix}$$

- What are the dimensions of the Jacobian matrix of f using (row, column) format?

- What is $\frac{\partial f_2}{\partial y}$?
- What is $\frac{\partial^2 f_1}{\partial x \partial y}$?

1.6 Push-through Identity (3 marks)

Suppose A is $m \times n$, B is $n \times m$, and the $m \times m$ matrix $(\lambda I + AB)$ is invertible. Prove that the $n \times n$ matrix $(\lambda I + BA)$ is invertible. *Hint.* Show that $(\lambda I + BA)x = 0$ implies $(\lambda I + AB)y = 0$, where $y = Ax$.

1.7 Regression Coefficients (3 marks)

In building a linear regression model for a particular data set, you observe the coefficient of one of the features having a relatively high negative value. Which one is true and why?

- This feature has a strong effect on the model (should be retained)
- This feature does not have a strong effect on the model (should be ignored)
- It is not possible to comment on the importance of this feature without additional information

1.8 Independent Features (6 marks)

Given n training data with d features, let the target value vector be $Y = \begin{bmatrix} y^{(1)} \\ \dots \\ y^{(n)} \end{bmatrix} \in \mathbb{R}^n$ and data samples

be $X = \begin{bmatrix} x_1^{(1)}, \dots, x_d^{(1)} \\ \dots \\ x_1^{(n)}, \dots, x_d^{(n)} \end{bmatrix} \in \mathbb{R}^{n \times d}$. In this context, x_j denotes the j^{th} column of this matrix. Show that if

we train a regressor on just one of the features (from d features), we then have $w_j = \frac{x_j^T y}{x_j^T x_j}$, where w_j is the j^{th} element of the parameter vector.

1.9 Ridge Regression (5 marks)

Consider $X \in \mathbb{R}^{n \times d}$ as the training data with n samples and d features, and $y \in \mathbb{R}^n$ as the labels. Using the squared L2-norm with an L2-regularization, the minimization problem becomes:

$$\arg \min_{w \in \mathbb{R}^d} \frac{1}{2} \|Xw - y\|^2 + \frac{\lambda}{2} \|w\|^2$$

Where λ is a tuning hyper-parameter. Setting the gradient to zero yields the following expression for the regression coefficients:

$$w = (X^T X + \lambda I_d)^{-1} (X^T y).$$

Is $(X^T X + \lambda I_d)$ always invertible? If yes, prove it. Otherwise, what conditions should X or λ have?

1.10 Multi-Class Classification with Softmax (3 marks)

Consider the dataset below, which has $n = 10$ training examples, $d = 2$ features, and $k = 3$ classes:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 0 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}, y = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 2 \\ 2 \\ 2 \\ 3 \\ 3 \\ 3 \end{bmatrix}$$

Suppose we fit a multi-class linear classifier using the softmax loss, and we obtain the following weight matrix:

$$W = \begin{bmatrix} +2 & -1 \\ +2 & -2 \\ +3 & -1 \end{bmatrix}$$

Under this model, what class label would we assign to $\hat{x} = [1, 1]$? (Show your work.)

1.11 Multi-Class Logistic Regression (6 marks)

Assuming that we have n training examples, d features, k classes, t testing examples, and T iterations of gradient descent for training:

- In $O()$ notation, what is the cost of training the softmax classifier with gradient descent?
- What is the cost of classifying the t test examples?

2 Coding Question (20 marks)

In this part, we will use linear regressors with the help of different basis functions. In this homework, you are meant to use the closed form of the optimal parameters with the Least Square Error loss function described in the class. You are not going to find the parameters iteratively.

Dataset: You are asked to work on the Boston House Prices dataset. This dataset consists of 506 data samples and 13 real attributes (features). The target value is the Median value ('MEDV') of owner-occupied homes in \$1000's (last column). The dataset is a csv file in the assignment1 folder.

Allowed packages: Pandas, Matplotlib, NumPy.

Deliverables: Please have the code, figures, and discussions in a Jupyter Notebook file (.ipynb). Your file should have five main sections, one for each task as follows:

1. First of all, we recommend checking whether the dataset includes missing parts. You can use Data Cleaning and Exploratory Data Analysis techniques to achieve a clean dataset. Then split the dataset into a train set (first 80% of the data) and a test set (last 20% of the data). Do NOT use the test set unless for final loss computation.
2. Plot the target values ('MEDV') with respect to the 'LSTAT' feature (first plot) and with respect to the 'DIS' feature (second plot). Can you see any dependencies between the target values and 'LSTAT' or 'DIS'?
3. Using the closed form of the linear regression for the Least Square Error, find the optimal regression parameters (weights). Plot the predicted values with respect to 'LSTAT' and 'DIS' in two separate plots. Can you see any dependencies between the predicted value and 'LSTAT' or 'DIS'?
4. We aim to include both the original features and the square of each feature in our data. This doubles the number of columns in the data matrix, increasing from 13 to 26 columns. Find the optimal parameters based on the new set of features and plot the predicted values with respect to 'LSTAT' and 'DIS' in two separate plots.
5. Report the MSE loss on training and testing sets for each section. Is adding the 2^{nd} order of features helpful?